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Customer Purchase Behavior Analysis

PREDICTIVE ANALYTICS SYSTEM

Analyze • Predict • Decide





PROJECT OVERVIEW

Project Summary
Using customer purchase data to analyze behavior and predict customer type (New vs Returning) to support smarter business decision-making.
Objectives:

- Analyze customer purchase data
- Predict customer type
- Generate business insights
- Enable data-driven decisions

Workflow Pipeline

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graph TD; A[Data Collection] --> B[Data Cleaning]; B --> C[Exploratory Data Analysis]; C --> D[Machine Learning]; D --> E[Prediction]; E --> F[Business Recommendations];
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A structured analytics pipeline transforms raw customer purchase data into actionable predictions and strategic business insights.

Raw customer purchase data was cleaned and preprocessed to ensure accuracy, consistency, and readiness for machine learning model training.

Removed missing data & null values
Eliminated duplicate records
Encoded categorical variables

Key Visualizations: Sales by Region (bar chart) • Profit by Category (bar chart)
• Customer Segment Distribution (donut chart) • Correlation Heatmap (color-coded matrix)

DATA PREPARATION & VISUALIZATION





Train-Test Split
80% Training
20% Testing
Feature Selection: Region,
Category, Sales, Profit, Segment,
Age, Gender, Discount

Best Model
Random Forest
Highest Accuracy among all
evaluated models with robust
performance on unseen data

Three classification models were evaluated on an 80/20 train-test split to
predict customer type. Feature selection ensured only the most relevant
variables were used, improving model accuracy and reducing overfitting.

MACHINE LEARNING MODEL

Logistic Regression
Decision Tree
Random Forest

Random Forest outperformed other models with the highest
classification accuracy, making it the selected model for the
customer type prediction system.

The prediction pipeline takes customer and purchase features as input and processes them through a trained machine learning model to classify the customer type — enabling smarter, data-driven business decisions.

PREDICTION SYSTEM

INPUT FEATURES

Region, Category, Product, Quantity, Sales, Profit, Customer Segment, Age, Gender, Discount Percentage

PREDICTION WORKFLOW

Input Features → Data Preprocessing → ML Model → Output Prediction → Business Action

Output Classes:

- New Customer
- Returning Customer



Key findings from the predictive analytics system reveal actionable patterns in customer purchase behavior, regional sales performance, and product profitability — enabling smarter, data-driven business decisions.

Strategic Recommendations:

- Focus marketing efforts on top-performing sales regions
- Increase inventory for best-selling product categories
- Launch customer loyalty programs targeting key segments
- Optimize discount strategies to maximize overall profit
- Leverage predictive analytics for forward-looking decisions

Highest Sales Region identified with strongest revenue contribution and customer volume.

Most Profitable Category and Top Selling Products highlighted for inventory prioritization.

Best Customer Segment determined
Discount Impact analyzed for profit optimization
Predictive model supports strategic planning

BUSINESS INSIGHTS & RECOMMENDATIONS



Technology Stack

Core tools and libraries used to build the Predictive Analytics System — from data processing to model deployment and version control.

Key Results Achieved

Successfully cleaned and preprocessed the customer dataset. Developed a high-accuracy predictive ML model. Extracted valuable business insights from data. Built an interactive prediction dashboard prototype for real-time use.



TOOLS & RESULTS

Technology Stack:
Python & Google Colab
Pandas, NumPy
Matplotlib, Seaborn
Scikit-learn, Joblib
Flask / Streamlit
GitHub

Model achieved strong predictive accuracy.
Dataset fully cleaned and preprocessed.
Business insights extracted and validated.
Interactive dashboard prototype successfully
built and deployed.



THANK YOU & FUTURE SCOPE

Achievements: Comprehensive data analysis & predictive modeling, actionable business insights, functional prediction system.

Future Scope: Real-time dashboard, cloud deployment, advanced ML models & continuous retraining.
Questions & feedback are welcome!